



Batterjee Medical College
For Science and Technology

Gastric Tumors

Discipline: Pathology
Block: M3-Digestion & Defense

Learning Outcomes



- By the end of this lecture, students must be able to
 1. Describe the etiology, pathogenesis, morphology and clinical features of gastric neoplasms

- Gastric cancer arises from malignant transformation of gastric cells
- Hereditary or Sporadic
- **Adenocarcinoma, MALT Lymphoma, Leiomyosarcoma, GIST, NET**
- **Adenocarcinoma:** Most common malignancy of the stomach
- Comprises **> 90%** of all gastric cancers

Separated morphologically into 2 types

1. Intestinal type, which tends to form **bulky masses**

2. Diffuse type, which **infiltrates** and **thickens** the **gastric wall**

- Early symptoms: **Dyspepsia, dysphagia, and nausea** (like chronic gastritis)
- These tumors are often discovered at advanced stages with late symptoms:
Weight loss, anorexia, altered bowel habits, anemia, and hemorrhage

Gastric Adenocarcinoma



- **Epidemiology:** Incidence varies markedly with geography
- Japan, Chile, Costa Rica, and Eastern Europe, the incidence is up to 20-fold higher than in North America, northern Europe, Africa, and Southeast Asia
- Increased incidence rates in patients aged between 55 and 80; rare in those under 50 years
- M:F = 2 to 3: 1

Gastric Adenocarcinoma



Etiology

- Most sporadic gastric cancers are usually related to *H. pylori* infection
- Chronic *H. pylori* infection is recognized as the **main cause** of non-cardia (distal) gastric cancer
- **Epstein Barr Virus (10%** of gastric cancers)
- **Smoking**
- Heavy **alcohol** consumption

Gastric Adenocarcinoma



Other proposed risk factors

- Diet (salt, preserved meats, coffee)
- Low socioeconomic status
- Obesity
- Pernicious anemia
- Previous gastric surgery
- Race / ethnicity
- Blood group A

Gastric Adenocarcinoma



Pathogenesis

- **Diffuse** gastric cancer
- Loss of function mutation in tumor suppressor gene **CDH1** mutations seen in 30% of hereditary diffuse gastric cancer and 50% of sporadic diffuse gastric cancer
- **CDH1**, encodes the cell adhesion protein **E-cadherin**
- **E-cadherin loss** is a key step in the development of **diffuse gastric cancer**
- Hereditary diffuse cancer: **α -Catenin** mutation
- Diffuse tumors are genomically stable and carry fewer mutations than intestinal-type gastric cancers

Pathogenesis

- **Intestinal-type gastric cancers** can be divided into molecular subtypes characterized by
 1. Chromosomal instability (CIN)
 2. Micro-satellite instability (MSI)
 3. Involvement of Epstein-Barr virus (EBV)
- **CIN**: Prevalent genes: **APC, SMAD 4, SMAD 2, TP53** ;
Site mostly involved: **Antrum** and **pyloric channel**
- **MSI** cancers favor the **gastric body** and typically display **MLH1** silencing
- **EBV** associated: **Fundus** and **proximal body**: silencing of CDKN2A, PIK3CA, increased PD-L1/2 expression

Gastric Adenocarcinoma: Morphology



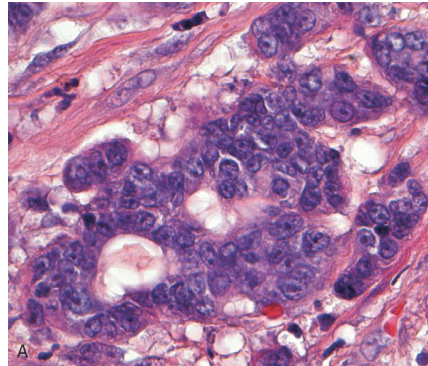
Intestinal type

- Most distal gastric adenocarcinomas occur in the gastric antrum
- Lesser curvature > Greater curvature
- Bulky tumors; composed of **glandular structures**; frequently grow along broad cohesive fronts
- Form exophytic mass or an ulcerated infiltrative tumor
- Tumor cells: contain apical mucin vacuoles, and abundant mucin may be present in gland lumina
- Extracellular mucin (Mucin lakes)

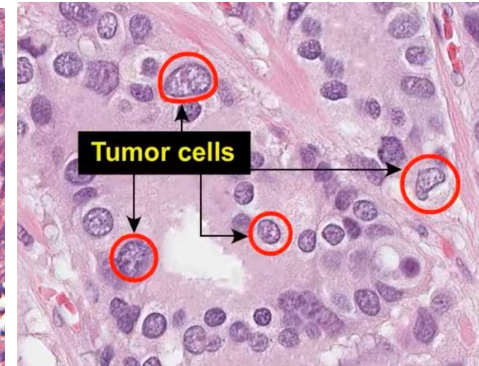
Gastric Adenocarcinoma: Morphology



Gastric adenocarcinoma. (A) Intestinal-type adenocarcinoma consisting of an elevated mass with heaped-up borders and central ulceration.



columnar, gland-forming cells
infiltrating through
desmoplastic stroma



Gastric Adenocarcinoma

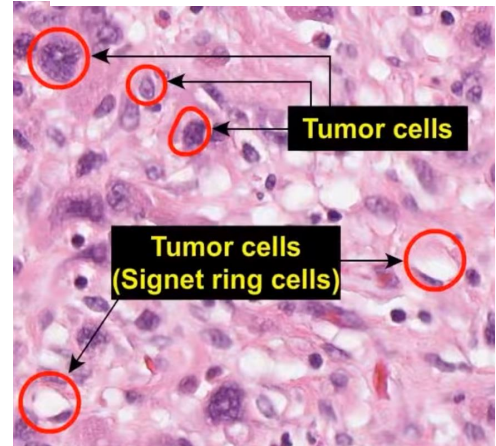
MORPHOLOGY

Diffuse type

- Difficult to appreciate a mass
- Infiltrative tumors often evoke a desmoplastic reaction; Stiffens the gastric wall.
- Diffuse flat rugae and rigid thick wall: When large areas of infiltration; Leather bottle stomach (**linitis plastica**)



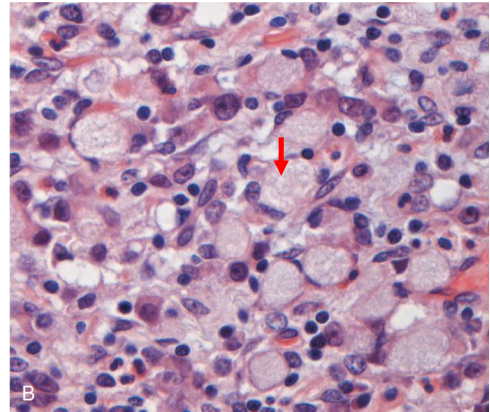
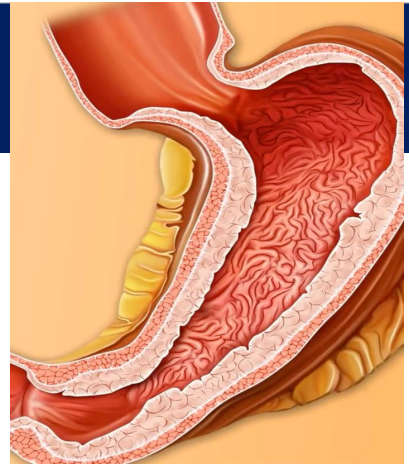
(B) Diffuse Infiltrative type (**linitis plastica**)



Gastric Adenocarcinoma: Morphology

Diffuse type

- Typically composed of signet-ring cells
- Small clusters and individual discohesive cells permeate the wall
- Cells have large mucin vacuoles with nucleus at periphery (signet ring cell);
- No glands.
- Mucin lakes



Gastric Adenocarcinoma



Clinical Features

Intestinal-type gastric cancer

- Predominates in high-risk areas
- Develops from precursor lesions including flat dysplasia and adenomas
- Mean age of presentation is 55 years, and the male-to-female ratio is 2 : 1.
- Diffuse type: Relatively uniform incidence across countries, and there are no identified precursor lesions

Clinical Features

- The **depth of invasion** and the **extent of nodal and distant metastases** remain the most powerful prognostic indicators in gastric cancer.
- Local invasion into the duodenum, pancreas, and retroperitoneum is common
- With surgical resection, the 5-year survival rate of early gastric cancer can exceed 90%, even if lymph node metastases are present.
- In contrast, the 5-year survival rate for advanced gastric cancer remains less than 20%

Gastric Adenocarcinoma



Metastases are often present at diagnosis

Sites most commonly involved include

- Supraclavicular sentinel lymph nodes (Virchow node)
- Periumbilical lymph nodes (Sister Mary Joseph nodule)
- Ovaries (Krukenberg tumor)
- The left axillary lymph node
- Pouch of Douglas.

Gastrointestinal stromal tumor (GIST)



- **Gastrointestinal stromal tumor (GIST)**
- MC mesenchymal tumor of the abdomen
- Can occur anywhere along the tubular GIT
- 50% of these occur in the stomach
- Arise from the interstitial cells of Cajal, or pacemaker cells
- Increased incidence with neurofibromatosis type 1.
- Diagnosed at a peak age of 60 years
- Symptoms of GISTs at presentation are due to mass effect

Gastrointestinal stromal tumor (GIST)



Pathogenesis

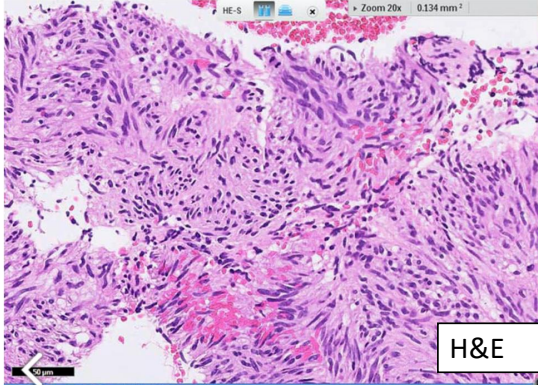
- 75% of all GISTs have oncogenic, gain of- function mutations in the **receptor tyrosine kinase KIT. (c-KIT)**
- Another 8% of GISTs have mutations in platelet-derived growth factor receptor α (PDGFRA).
- Tumors with mutations in KIT or PDGFRA often respond to the tyrosine kinase inhibitor **imatinib**, whereas tumors without these mutations are generally resistant.

Gastrointestinal stromal tumor (GIST)



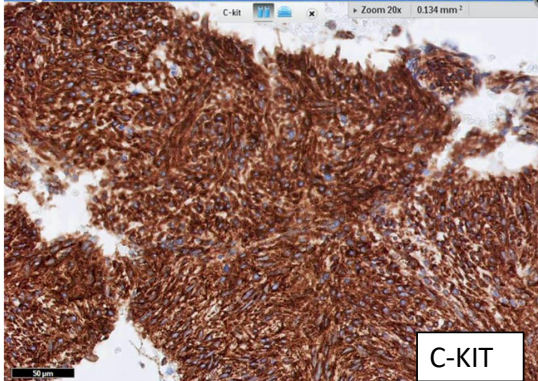
- Size: Can reach upto 30 cm in diameter.
- Solitary, well-circumscribed, fleshy mass with ulcerated overlying mucosa
- Cut surface: Whorled appearance.
- Metastases: Form serosal nodules throughout peritoneal cavity or within the liver
- GISTs composed of thin elongated cells are classified as **spindle cell type**
- GISTs dominated by plumper, epithelial-appearing cells are termed **epithelioid type**
- Most useful IHC marker: **KIT (CD117)**

Gastrointestinal stromal tumor (GIST)

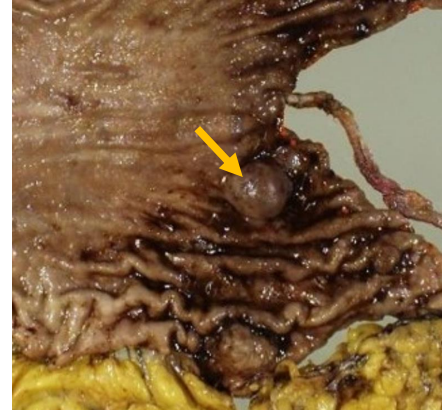


GIST:
Spindle
type

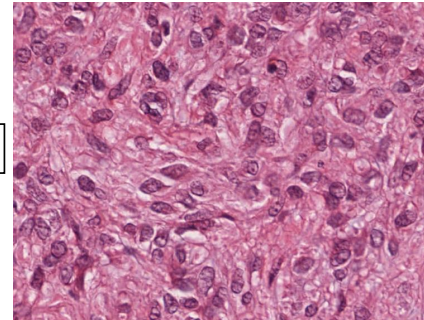
H&E



C-KIT



GIST in a
gastrectomy
specimen



Epithelioid type

Neuroendocrine tumors (Gastrointestinal carcinoid)



- Neuroendocrine tumours (NETs) start in cells of the neuroendocrine system.
- Neuroendocrine cells are spread throughout the body

Carcinoid (Well differentiated neuroendocrine tumor)

- Called “Carcinoid” because they are slowly growing than carcinomas
- The most current WHO classification describes these as low- or intermediate-grade neuroendocrine tumors
- For High grade = Neuroendocrine carcinoma

Carcinoid (Well differentiated NET)



Most are found in the GI tract, and more than 40% occur in the small intestine.

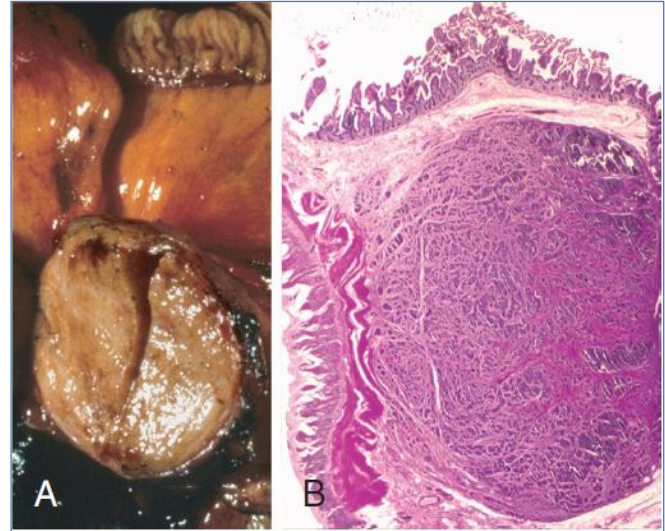
Other sites: Tracheobronchial tree and lungs

Gastric neuroendocrine neoplasms may be associated with

- Endocrine cell hyperplasia
- Autoimmune chronic atrophic gastritis
- MEN1
- Zollinger-Ellison syndrome

Carcinoid: Morphology

- Gross: Polypoid intramural or submucosal mass
- In stomach: Arise within oxyntic mucosa.
- Overlying mucosa may be intact or ulcerated
- In intestines: Tumors may invade deeply to involve the mesentery.
- Yellow or tan in color
- Firm as a consequence of intense desmoplasia; may cause obstruction of the small bowel.



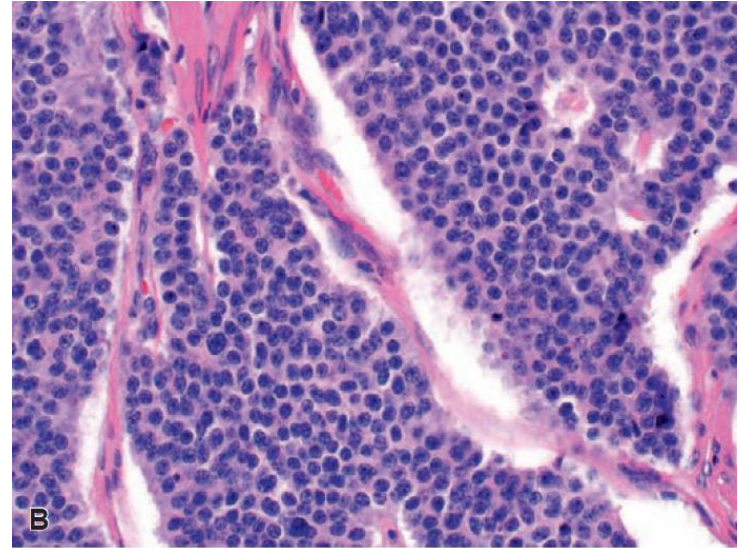
Gross cross section of a submucosal tumor nodule. (B) Microscopically the nodule is composed of tumor cells embedded in dense fibrous tissue.

Carcinoid: Morphology

Histology

Islands, trabeculae, strands, glands, and sheets of uniform cells with scant, pink granular cytoplasm and a round to oval nucleus with a “salt and pepper” chromatin pattern

Positive for endocrine granule markers such as synaptophysin and chromogranin A



Nuclei are round with a finely speckled chromatin pattern and indistinct nucleoli.

Clinical features

- MC: 6th decade; May appear at any age
- Symptoms reflect hormones released by the tumor cells
- Those that produce gastrin cause Zollinger-Ellison syndrome.

Carcinoid syndrome: Develops in fewer than 10% of patients

- Caused by vasoactive substances secreted by the tumor into the systemic circulation.
- Lead to cutaneous flushing, sweating, bronchospasm, colicky abdominal pain, diarrhea, and right-sided cardiac valvular fibrosis.
- Cyanosis of the face and anterior part of the chest, intermittent hypertension, palpitation

Clinical features

- Foregut neuroendocrine tumors within the stomach and duodenum rarely metastasize and are generally cured by resection
- Midgut neuroendocrine tumors arise in the jejunum and ileum, are often multiple, and **tend to be aggressive**
- Hindgut neuroendocrine tumors arising in the appendix and colorectum are almost always benign

Summary



- **Gastric adenocarcinoma** is the most common gastric malignancy, accounting for **>90% of gastric cancers**.
- Gastric adenocarcinoma is divided into **intestinal and diffuse types**, with distinct morphologic features.
- **H. pylori infection** is the main cause of non-cardia gastric cancer; other associations include **EBV, smoking and heavy alcohol consumption**.
- **Intestinal-type adenocarcinoma** is gland-forming, whereas the **diffuse type** shows discohesive signet-ring cells and may produce **linitis plastica**.
- **Depth of invasion and extent of nodal and distant metastases** are the most powerful prognostic indicators in gastric cancer.
- **GIST** arises from interstitial cells of Cajal, commonly involves the stomach, and is frequently associated with **KIT mutations**.
- **GISTs** may show spindle or epithelioid morphology, with **KIT (CD117)** being the most useful IHC marker.
- **Neuroendocrine tumors** show characteristic “salt-and-pepper” chromatin and are positive for **synaptophysin and chromogranin A**

Case Study



- A 53-year-old woman has had nausea, vomiting, and mid epigastric pain for 5 months. On physical examination, there are no significant findings. An upper gastrointestinal radiographic series shows gastric outlet obstruction. Upper gastrointestinal endoscopy shows an ulcerated mass that is 2 × 4 cm at the pylorus. Which of the following neoplasms is most likely to be seen in a biopsy specimen of this mass?
- (A) Non-Hodgkin lymphoma
 - (B) Neuroendocrine carcinoma
 - (C) Squamous cell carcinoma
 - (D) Adenocarcinoma
 - (E) Leiomyosarcoma

References



1. Kumar, V., Abbas, A. K., Aster, J. C., Debnath, J., & Das, A. (Eds.). (2026). *Robbins, Cotran & Kumar pathologic basis of disease* (11th ed.). [Elsevier](#)
2. AMBOSS SE. (n.d.). *AMBOSS: Medical knowledge platform for doctors and students*. Retrieved August 29, 2026, from [AMBOSS](#)

Thank You